

Impact of Second Dose Vaccination Rates on Measles Incidence in African Regions

S.G. Chowdhary

AT3 Scientific Report

[SLE251, Research Methods & Data Analysis]

Introduction

Measles, caused by the Morbillivirus, is a highly contagious viral disease characterized by symptoms of high fever, cough, runny nose, inflamed eyes, and a distinctive red rash. Transmission occurs via respiratory droplets from coughs and sneezes, making outbreaks rapid and widespread in unvaccinated populations (Pruthi, 2022; Shimizu, 2024; Perry & Halsey, 2004). The MMR (Measles, Mumps, and Rubella) vaccine plays a crucial role in preventing measles, as it induces immunity in over 95% of recipients after a single dose, with a second dose ensuring immunity in virtually all individuals. High vaccination coverage is essential to achieve herd immunity and interrupt transmission (Bester, 2016; Pruthi, 2022). African regions are critical for this study due to disparities in healthcare infrastructure and vaccination coverage. Despite global efforts, measles outbreaks remain frequent in these regions, exacerbated by logistical challenges, political instability, and misinformation (WHO, 2023).

Understanding the relationship between vaccination percentage and measles incidence in these areas is vital for informing public health strategies. This study aims to analyze the correlation between the incidence of measles relative to second dose vaccination percentage in WHO-designated African regions between 2000-2022. It is hypothesized that higher vaccination rates will correlate with reduced measles incidence. The null hypothesis of this report is that no significant correlation exists between the vaccination rates and the incidence of measles.

Method

Primary Data Resource

World Health Organization (WHO) (18 May 2024) 'African Region Reported cases of vaccine-preventable diseases (VPDs)', *World Health Organisation (WHO) International*, accessed 19 May 2024. <https://immunizationdata.who.int/dashboard/regions/african-region>

The data utilized in this study was obtained from the World Health Organization (WHO) Immunization Data Portal, specifically sourced from the WHO/UNICEF Estimates of National Immunization Coverage (WUENIC). Table 1 depicts yearly records of second dose vaccination coverage percentages and incidence of measles (per 100,000 of total population) for WHO-designated African regions from 2000 to 2022.

Table 1. Yearly Records of Second Dose Vaccination Coverage and Measles Incidence in WHO-Designated African Regions (2000-2022)

Year	Vaccination Percentage of Population, 2nd Dose	Incidence of Measles (per 100,000 of total population)
2022	45	8.16
2021	41	8.19
2020	40	10.63
2019	33	55.98
2018	25	11.66
2017	25	6.92
2016	22	3.65
2015	17	5.3
2014	10	7.62
2013	7	18.13
2012	6	12.34
2011	5	21.88
2010	5	23.15
2009	5	9.97
2008	4	4.54
2007	4	9.63
2006	4	12.87
2005	4	43.28
2004	4	30.05
2003	5	56.56
2002	5	41.52
2001	5	73.33
2000	5	83.23

The dataset comprises yearly records of measles incidence (per 100,000 total population) and second dose vaccination coverage percentages. Initially, the measles incidence data was presented per 1,000,000 total population on the WHO website. For more suitable analysis in R, the data was converted to per 100,000 total population. The selection criteria focused on the completeness and relevance of the data to ensure a robust analysis. The dataset was chosen for its comprehensive coverage of the target region and the consistency of data reporting over the years.

The focus of this report has been allocated to the second dose vaccination coverage, as the second dose plays a critical role in ensuring immunity. While the first dose of the MMR vaccine induces immunity in over 95% of recipients, the second dose ensures immunity in virtually all individuals (Bester, 2016; Pruthi, 2022). This second dose is vital for achieving and maintaining herd immunity, particularly in regions with significant healthcare disparities.

Statistical Analysis

To analyze the relationship between vaccination rates and measles incidence, linear regression and correlation analysis will be employed. This method is appropriate for examining the correlation between two continuous variables: vaccination coverage percentage and measles incidence. The following steps were conducted utilizing the R-Commander programming software (Fox and Bouchet-Valat, 2024).

1. A scatterplot was generated in order to determine that assumptions of skew, symmetry and trend were met, the annual percentage of vaccination was the x-variable and the incidence of measles was used as the y-variable.
2. After reviewing the scatter plot with the attached box plot, visual outliers were identified within the data and the initial scatterplot exhibited a positively skewed distribution.
3. In order to enhance the data's reliability and symmetry, a log10 transformation of the y-variable was performed, followed by the generation of another scatterplot.
4. After the transformation, both vaccination percentages and incidence rates exhibited greater symmetry. However, a linear relationship was not evident and only a limited range of data displayed a negative linear trend. As there were no zero values in the dataset, a further data transformation was deemed unnecessary.
5. Utilizing the transformed data, a correlation test and linear regression were conducted to derive the nature of the relationship between vaccination coverage and measles incidence. The correlation test quantified the degree of association, while the linear regression provided insights into the strength and direction of the relationship.

Results

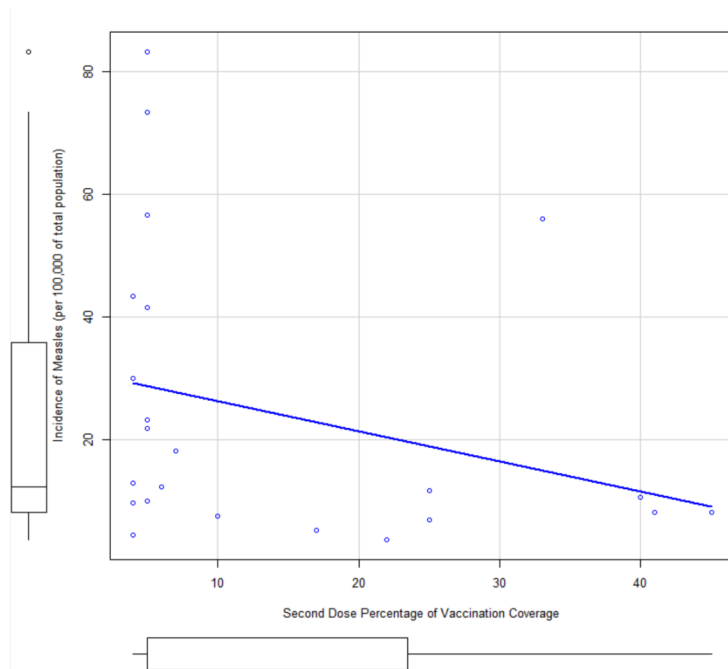


Figure 1. Scatterplot of Measles of Incidence (per 100,000 of total population) vs. Second Dose Percentage of Vaccination Coverage. The least squares regression line is shown.

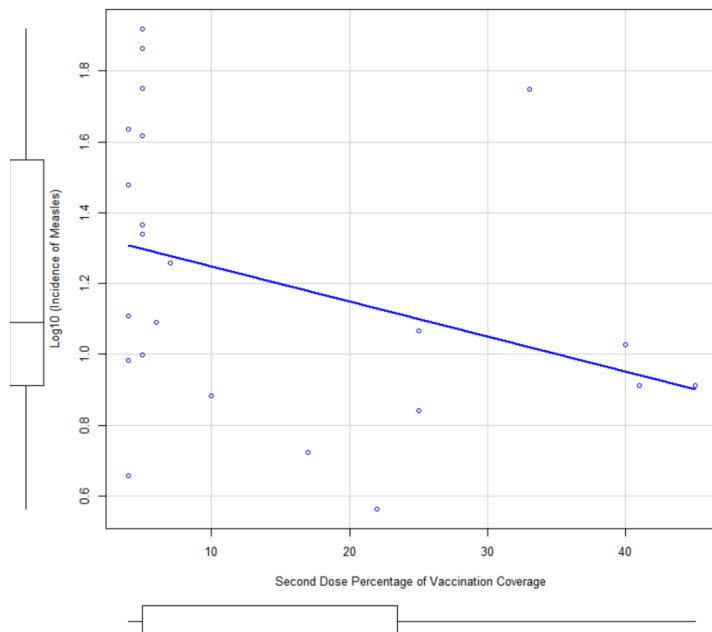


Figure 2. Scatterplot of Log10 Transformed Measles of Incidence (per 100,000 of total population) vs. Second Dose Percentage of Vaccination Coverage. The least squares regression line is shown.

Figure 1 displays a scatterplot of measles incidence (per 100,000 of total population) versus the second dose percentage of vaccination coverage. Figure 2 presents a scatterplot of the log10 transformed measles incidence against the second dose percentage of vaccination coverage. The transformation in Figure 2 was performed to address the skewness observed in the original data and to improve the reliability and symmetry of the distribution.

Correlation Test

The correlation test highlighted a potential negative correlation between vaccination rates and measles incidence. The results of the linear regression analysis were as follows; $t = 1.6647$, $df = 21$, $p\text{-value} = 0.1108$. Since the $p\text{-value}$ of 0.1412 is greater than the standard significance level of 0.05, the results are non-significant. This means that the vaccination coverage (independent variable) has no significant relationship with measles incidence (dependent variable) in the dataset. Hence, the null hypothesis, which states that no significant correlation exists between vaccination rates and measles incidence, must be accepted.

Linear Regression

The linear regression analysis yielded a multiple r^2 value of 0.1166, indicating that approximately 11.66% of the variation in measles incidence can be explained by vaccination coverage. A low r^2 value suggests that the model does not explain much of the variability in the data, implying that other factors are likely contributing to changes in measles incidence. The regression equation derived from the analysis is:

$$\log_{10}(\text{measles incidence}) = 28.431 + (-11.817) \times (\text{vaccination coverage})$$

The regression coefficients are as follows: $t = -1.665$, intercept = 28.431 (std error = 9.005), and slope = -11.817 (std error = 7.098). The negative slope suggests a potential inverse relationship between vaccination coverage and measles incidence; meaning that as vaccination coverage (independent variable) increases, measles incidence (dependent variable) should decrease. However, due to the non-significant $p\text{-value}$, this relationship cannot be confirmed.

Discussion

This study aimed to analyze the correlation between second dose vaccination rates and measles incidence in WHO-designated African regions from 2000 to 2022. The hypothesis asserted that higher vaccination rates would correlate with reduced measles incidence, while the null hypothesis stated that no significant correlation exists. The results of the linear regression analysis indicated a potential negative correlation; however, this relationship was not statistically significant. Hence, the null hypothesis was accepted, suggesting that vaccination rates did not significantly impact measles incidence within the analyzed dataset.

The scatter plots provided a visual representation of the data. Figure 1 displayed a negatively skewed distribution, prompting a log₁₀ transformation of the y-variable to improve symmetry and reliability in Figure 2. Despite this transformation, the correlation test and subsequent linear regression did not reveal a significant relationship. The multiple r-squared value of 0.1166 indicated that only 11.66% of the variation in measles incidence could be explained by vaccination coverage. The negative slope in the regression equation suggested an inverse relationship, but the lack of statistical significance prevented definitive conclusions.

These findings are consistent with other studies highlighting the complex interplay of factors influencing measles incidence beyond vaccination coverage alone. For instance, Afolabi and Ilesanmi (2021), Bagcchi (2021), and Neely et al. (2022) noted that logistical challenges, political instability, and misinformation significantly affect vaccination coverage and disease control in African regions. While the dataset was comprehensive, it did not account for all variables impacting measles incidence, such as cultural beliefs and practices, healthcare access, and social factors (e.g., political stability, media influence, population density). Additionally, data accuracy and reporting consistency could vary across African regions and years (Bagcchi, 2021). Future research should consider a more targeted approach, examining individual countries or sub-regions within Africa to capture localized variations and influences. Utilizing qualitative data on healthcare infrastructure, community engagement, and education could provide deeper insights.

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